

PROFILE

Jinmin Kim

3D Registration and Robot Software Engineer

An R&D engineer whose work spans medical navigation, XR, 3D registration, and multi-sensor integration. I connect coordinate models and algorithms to physical devices, user interfaces, and explicit validation flows.

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EXPERIENCE AND EDUCATION

2023-02 - Present	DIGITRACK Inc.	Robot Software Engineer Implement medical navigation, XR, 3D registration, multi-sensor perception, industrial system integration, and field validation.
2021-03 - 2023-02	DGIST	M.S., Robotics and Mechatronics Engineering Researched mandibular fracture reduction optimization, surgical navigation, and XR.
2020-07	UNIST U-SURF	Summer Research Intern Reviewed soft-robotics literature and conducted magnetic-force experiments.
2015-03 - 2021-02	Kumoh National Institute of Technology	B.S., Mechanical System Engineering Completed undergraduate system-vision research and invention activities.

USAGE-BASED ENGINEERING EXPERIENCE

3D Registration and Optimization Used PCA, ICP, CLPSO, Open3D, OpenCV, and SciPy for feature, surface, and coordinate-system problems.	Medical Navigation and XR Built working applications with 3D Slicer, VTK, Qt, OpenIGTLink, Unity, MRTK, Meta SDK, PUN2, and Photon Voice.
Robot and Sensor Integration Integrated ROS 2, Unity, Isaac Sim, NDI tracking, HoloLens 2, Meta Quest, RGB, ToF, and LiDAR.	Application-layer Engineering Used Python, C#, and C++ for algorithms, XR applications, APIs, ROS 2 packages, and device interfaces.

Research and Public Evidence

2024 / JOINT FIRST AUTHOR

A Proof of Concept: Optimized Jawbone-Reduction Model for Mandibular Fracture Surgery

Journal of Imaging Informatics in Medicine, Q1 SCIE

[Public article](#)

2022 / INTERNATIONAL CONFERENCE PRESENTATION

Dental Occlusion Model Using Arch Line for Mandibular Fracture Surgery

ACCAS 2022

ACHIEVEMENT SIGNALS

PATENTS

7 applications / 3 grants

AWARDS

9 total

LANGUAGES

Korean - Native / Japanese - JLPT N2

SELECTED AWARDS

2024 Best Poster Award, Korean Society of Medical Metaverse

2023 Best Paper Award, Korean Society of Medical Robotics

2019 Grand Prize, International TRIZ Competition

REPRESENTATIVE WORK SCOPE

Surgical Navigation / Mandibular Fracture Optimization / Life Careverse / rTMS Navigation / Surface-guided Respiratory Tracking / SKADI Software / Autonomous Forklift Registration / AI Build Lab

Two-page CV using only public facts approved on 2026-08-16.